

Formation of a mutagenic heterocyclic aromatic amine from creatinine in urine of meat eaters and vegetarians.

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Liquid chromatography electrospray ionization mass spectrometry (MS) with a triple quadrupole MS was used to identify known and novel heterocyclic aromatic amines (HAAs) in human urine. The identities of 2-amino-3,8-dimethylimidazo[4,5-f]quinoxaline (8-MeIQx) and 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) were confirmed by their product ion spectra. The constant neutral loss scan mode was employed to probe for other analytes in urine that display the transition $[M+H]^{+} \rightarrow [M+H-CH_3]^{+}$, which is common to HAAs containing an N-methylimidazo moiety, and led to the detection of a previously unreported isomer of 8-MeIQx [Holland, R., et al. (2004) Chem. Res. Toxicol. 17, 1121-1136]. We now report the identification of another novel HAA, 2-amino-1-methylimidazo[4,5-b]quinoline (IQ[4,5-b]), an isomer of the powerful animal carcinogen 2-amino-3-methylimidazo[4,5-f]quinoline (IQ). The amounts of IQ[4,5-b] measured in the urine of human volunteers who consumed grilled beef ranged from 15 to 135% of the ingested dose, while the amounts of 8-MeIQx and PhIP excreted in urine were on average <2% of the ingested dose. Base treatment of urine at 70 degrees C increased the concentrations of 8-MeIQx and PhIP by as much as 6-fold, indicating the presence of phase II conjugates; however, the amount of IQ[4,5-b] increased by more than 100-fold. IQ[4,5-b] was also detected in the urine of vegetarians following base hydrolysis. The formation of IQ[4,5-b], but not IQ, 8-MeIQx, or PhIP, also occurred in urine incubated at 37 degrees C. Creatinine and 2-aminobenzaldehyde are likely precursors of IQ[4,5-b]. The detection of IQ[4,5-b] in the urine of both meat eaters and vegetarians suggests that this HAA may be present in nonmeat staples or that IQ[4,5-b] formation may occur endogenously within the urinary bladder or other biological fluids.